Path-ILC with a learned correction layer for high-accuracy industrial robots

A flexible-joint KUKA iiwa14 that self-learns its drivetrain error from joint-side encoders, then *generalizes* the correction with a small neural-network layer

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Background & motivation

Industrial robots are accurate *relative* to themselves but their *absolute* accuracy is limited by unmodeled **drivetrain error**: joint elasticity, gear/cycloidal transmission error, friction, backlash and thermal drift. Schwegel & Kugi (ICRA 2024) cancel this with a **path-parameter ILC** but need an external **laser tracker** to learn, and list open problems: speed variation, **contact**, a **model-based learning filter**, and **generalizing** the correction to **new paths**. **The robot-arm model I assume**

Each axis = a **two-mass flexible joint**: motor inertia J_m and link inertia J_ℓ joined by a torsional **spring** K **+ damper** d. The controller plans on the *rigid* model; the plant deflects, so $\Delta = \theta_m - \theta_\ell$ is a real, pose/load-dependent transmission error.

- $K \approx$18000→3000 Nm/rad, $d \approx$50→8 (root→wrist)
- $f_n \approx$10–195 Hz, $\zeta \approx$0.1–1.6
- Gear TE 0.3–0.8 mrad, 13–37 cyc/rev; backlash 0.5–1.2 mrad
- Stribeck F_s/F_c=1.7; thermal k_T 3–6 mrad/T; 19-bit encoder

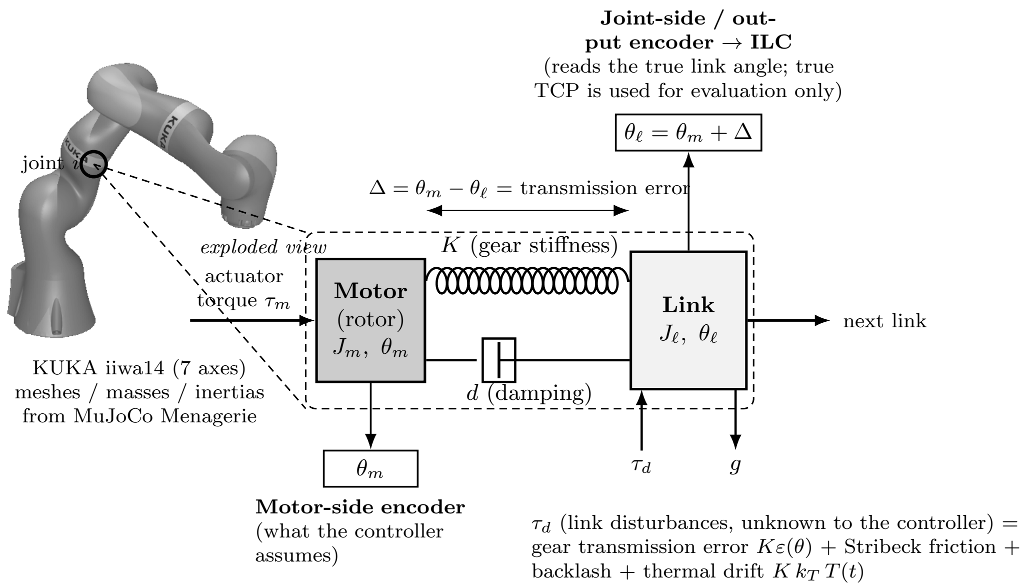
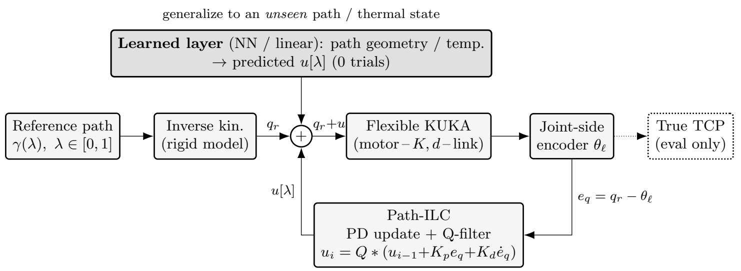


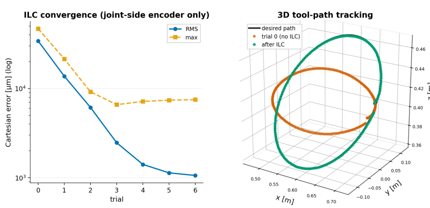
Fig. 1. Drivetrain model. One axis of the real iiwa14 is exploded into motor J_m – spring K/damper d – link J_ℓ, with the link disturbance torque τ_d and the two encoders. The **joint-side (output) encoder** feeds the ILC; the true TCP is evaluation only.

Method



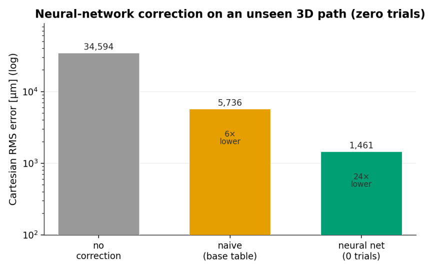
The **path-ILC** learns a table $u[\lambda]$ indexed by the path parameter λ (not time) with a PD update + Gaussian **Q-filter**; λ-indexing transfers across speeds. A small **neural-network layer** predicts $u[\lambda]$ for an *unseen* path / thermal state with **zero trials**.

Results (Cartesian RMS at the tool tip; simulation, seeded/reproducible)



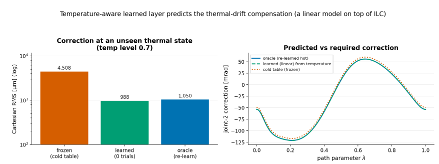
Convergence (encoder only): **34156→1061 μm, ~32×**

self-learned without a laser tracker



Unseen path, 0 trials: 34594→5736→**1461** μm

instant vs. re-running ILC trials



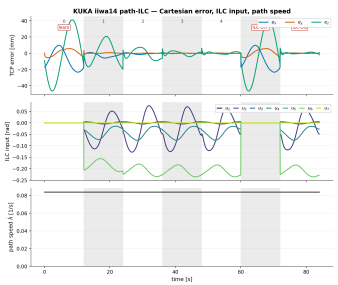
Unseen thermal state: frozen 4508→**988**≈oracle 1050 μm

a fixed table cannot track drift; this can

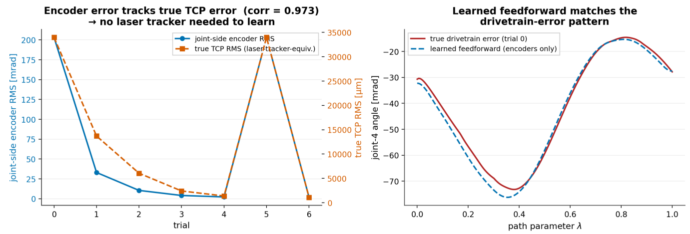
Honest scope. Simulation. The joint-side (output) encoder *directly* observes the link, so learning-without-a-tracker here is a *stronger* sensing assumption than the paper's motor-encoder-only robot; real secondary encoders have noise/resolution limits and a joint→task kinematic-calibration gap remains. Speed transfer is partial. **Reproduce all figures:** `python src/run_all.py`.

Appendix — analysis (a–d) & realistic-effects (e–i) figures

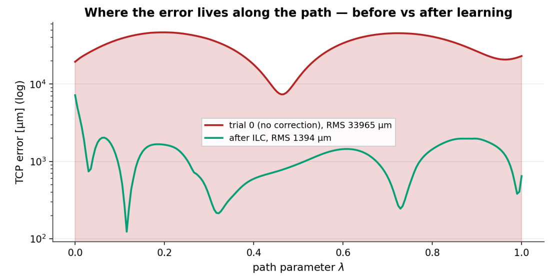
Supporting evidence behind the headline results — one panel per claim (cf. `FIGURES.md`). Cartesian RMS at the tool tip; simulation, seeded.



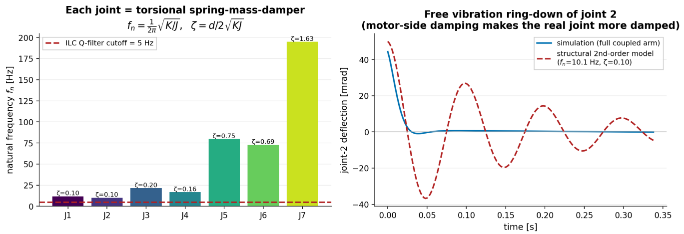
(a) — paper Fig. 5 reproduction: TCP error / ILC inputs / path speed over 7 trials (trial 5 ILC-*off*, 6 *on*).



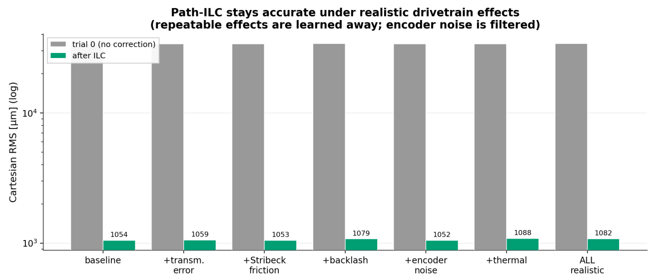
(b) — no-tracker validation: joint-side encoder RMS vs true TCP RMS in lock-step (corr 0.97).



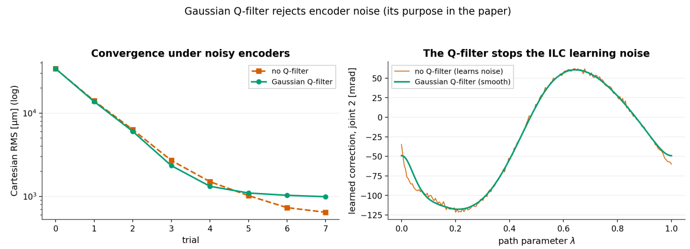
(c) — where the error lives along the path, before vs after learning (log scale).



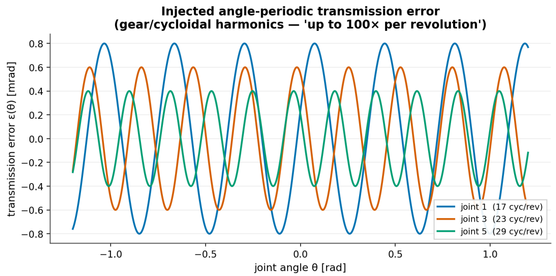
(d) — mechanical view: each joint a torsional spring-mass-damper; f_n 10–195 Hz vs the Q-filter cutoff.



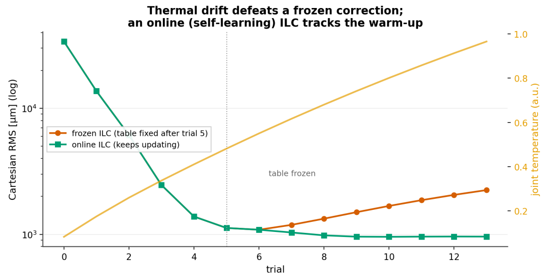
(e) — ablation: ILC stays ∼1 mm under every realistic effect (repeatable learned, noise filtered).



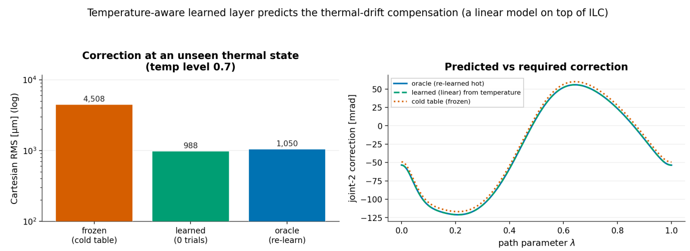
(f) — the Gaussian Q-filter rejects encoder noise (the learned correction is jagged without it).



(g) — the injected angle-periodic transmission error (gear/cycloidal harmonics, 13–37 cyc/rev).



(h) — thermal drift defeats a frozen ILC table; an online ILC tracks the warm-up.



(i) — temperature-aware learned model predicts the thermal compensation at an *unseen* state, 0 trials.